

SELF-CALIBRATING DRUG DELIVERY SYSTEMS CAPABLE OF ADAPTIVE RELEASE THRESHOLD ADJUSTMENT AFTER INITIAL TISSUE EXPOSURE

Abstract

Conventional drug delivery systems commonly operate with fixed formulations, predefined release of kinetics, or stimulus-response thresholds established before administration. These approaches can be effective but may perform poorly when the biological environment changes over time or differs substantially among patients. Self-calibrating drug delivery systems represent a proposed next stage in adaptive therapeutic delivery in which the system uses information obtained after tissue exposure to estimate its operating baseline and modify subsequent release behavior. This review develops a systems-level framework that distinguishes conventional controlled release, stimuli-responsive delivery, feedback-controlled delivery, closed-loop delivery, and genuinely self-calibrating delivery. Attention is given to the initialization phase after administration or implantation, during which protein adsorption, immune interactions, local biochemical gradients, biofouling and sensor drift can alter the relationship between a measured signal and the desired therapeutic response. We examine sensing modalities, signal processing, adaptive control, smart materials, actuation mechanisms, pharmacokinetic–pharmacodynamic modelling, quantitative performance metrics, and experimental validation strategies. Evidence across glucose-responsive systems, tumor-microenvironment-responsive delivery, inflammation-responsive materials and bioelectronic or neuromodulator platforms is considered with emphasis on whether published systems demonstrate true post-exposure calibration rather than simple stimulus responsiveness. Finally, we propose a maturity framework for classifying systems according to sensing, baseline estimation, threshold adaptation, feedback, and longitudinal recalibration. The major translational barriers are sensor drift, biofouling, biological heterogeneity, energy and communication requirements, manufacturing reproducibility, algorithmic validation, cybersecurity, regulatory oversight and the absence of standardized benchmarks for adaptive performance. The central proposition is that post-exposure calibration should be treated as a measurable design function rather than a broad synonym for intelligent drug delivery. This distinction can help organize the field and identify which technologies are genuinely moving toward autonomous, patient-adaptive therapy.

Keywords: Self-calibrating drug delivery; adaptive release; closed-loop drug delivery; stimuli-responsive materials; biosensors; tissue microenvironment; adaptive control; PK/PD modelling; precision therapeutics; intelligent drug delivery

1. Introduction

Drug delivery has progressed from passive dosage forms toward systems capable of controlling where, when and under what conditions, a therapeutic agent is released. Conventional oral, parenteral, transdermal, inhaled and implantable systems remain clinically important, but many rely on formulation properties or dosing schedules established before administration.

Physiological variability can alter absorption, distribution, metabolism, and elimination, while disease progression can change the local environment in which a delivery system operates. Consequently, the same formulation or release threshold may not produce the same therapeutic outcome across individuals or across time within a single patient.

Stimuli-responsive systems address part of this problem by linking drug release to endogenous or exogenous cues such as pH, temperature, glucose, enzymes, redox state, light, ultrasound, magnetic fields or electrical signals. Closed-loop platforms extend this concept by combining sensing, decision-making, and actuation. Reviews of biosensor-integrated delivery systems have described platforms in which biological signals are measured continuously and used to trigger or modulate drug release. Self-regulated devices have likewise been proposed to mimic physiological feedback and provide on-demand therapy.

However, responsiveness alone does not necessarily imply calibration. A system can respond repeatedly to a biomarker while retaining the same response threshold throughout its operation. The distinction becomes important in complex tissues because implantation or injection can change the local environment. Protein adsorption, inflammatory responses, fibrosis, changing oxygenation and biofouling can alter sensor output, material properties, and mass transport. A system that does not account for these changes may gradually deviate from its intended operating point even if its nominal stimulus-response mechanism remains intact.

Recent work has broadened the field toward intelligent and miniaturized drug-delivery devices that integrate biotechnology, electronics, materials science and artificial intelligence with real-time sensing and adaptive control. That development makes it increasingly important to define what is meant by adaptation and, more specifically, by self-calibration. In this review, self-calibration is used in a deliberately narrow sense: the delivery platform acquires information after exposure to the biological environment, estimates a relevant baseline or system state, and uses that information to modify one or more parameters governing subsequent therapeutic release.

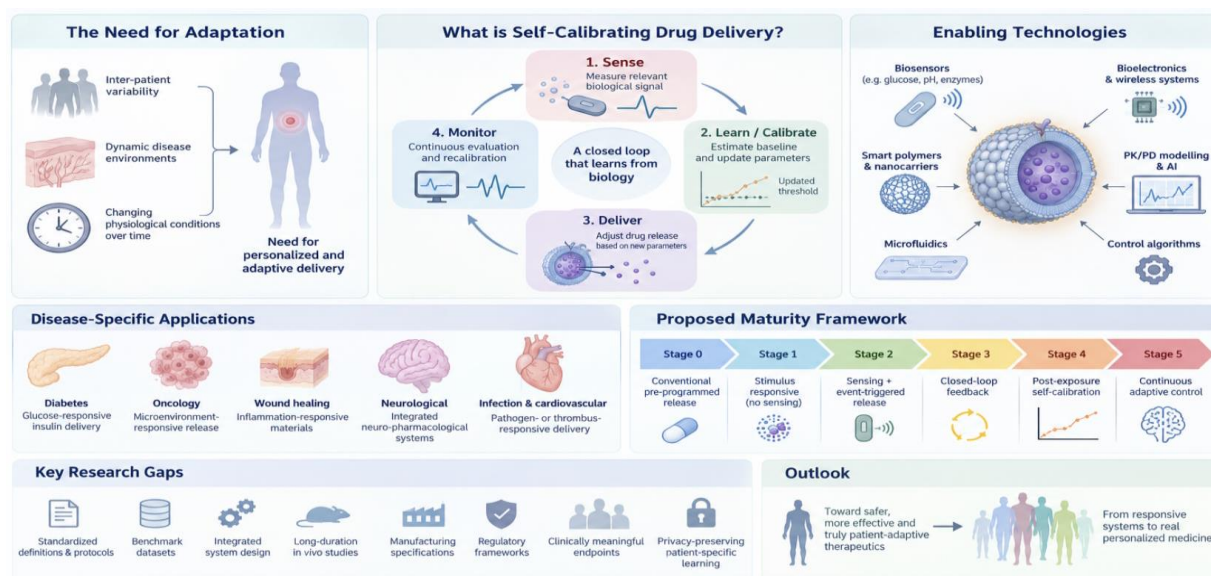


Fig. 1.

Example of graphical abstracts.

2. From Conventional Release to Self-Calibration

2.1 Conventional controlled-release systems

Conventional controlled-release drug delivery systems are mainly designed based on predefined formulation and drug-related properties. During development, factors such as drug solubility, polymer characteristics, membrane permeability, dosage-form geometry, and drug–excipient interactions are considered to obtain a desired release profile. Depending on the formulation design, drug release can be controlled through diffusion, dissolution, osmotic pressure, polymer erosion or degradation, and changes in the physical structure of the delivery system. By adjusting these parameters, formulations can be developed to provide approximately zero-order, first-order, or other predetermined release kinetics. These systems have been widely investigated and remain important in pharmaceutical development because their mechanisms are relatively well understood and can be evaluated using established *in vitro* and *in vivo* methods. Their manufacturing processes can also be standardized, which makes them easier to reproduce and characterize. From a regulatory perspective, predictable behaviour and well-defined formulation parameters are important advantages. However, conventional controlled-release systems generally operate according to a release program that is established before administration. Once the formulation is exposed to the patient's biological environment, factors such as gastrointestinal conditions, tissue characteristics, enzymatic activity, metabolism, disease-related changes, and individual variability may influence drug release and drug response. The formulation itself usually does not have the ability to recognize these changes and modify its release behaviour accordingly. Therefore, although conventional systems can provide controlled delivery, their control is primarily pre-programmed rather than adaptive. This limitation has encouraged the development of delivery approaches that can respond to biological conditions and, more recently, learn from those conditions.

2.2 Stimuli-responsive drug delivery

Stimuli-responsive drug delivery systems represent an important step beyond conventional controlled-release formulations because they allow drug release to change in response to a particular physical, chemical, or biological stimulus. In these systems, the delivery material is designed to undergo a specific change when it encounters a predefined trigger. Depending on the design, the material may swell, contract, degrade, change its permeability, undergo a phase transition, or alter its molecular interactions, resulting in a corresponding change in drug release. A wide range of stimuli have been investigated for this purpose. These include endogenous signals such as glucose concentration, pH, enzymes, reactive oxygen species, and redox conditions, as well as externally applied stimuli such as temperature, light, magnetic fields, and electrical signals. Examples include glucose-responsive systems for insulin delivery, pH-responsive carriers, enzyme-cleavable drug carriers, redox-responsive materials, and thermos responsive hydrogels. Such systems can provide greater biological responsiveness than conventional formulations because the release process is linked to a condition that may vary within the body. Despite this advantage, most stimuli-responsive systems still depend on a response mechanism that is predetermined during formulation development. In other words, the material is programmed to

respond to a particular stimulus according to a predefined relationship. For example, a specific glucose concentration may produce a particular change in permeability or drug release. If the biological environment differs substantially from the conditions anticipated during development, the magnitude or timing of the response may not be optimal. Thus, stimuli-responsive delivery should not automatically be considered self-calibrating. The stimulus may modify drug release, but the underlying relationship between the stimulus and the response can remain fixed. A system becomes more adaptive when it can not only detect a biological change but also estimate how it should respond under the individual's current conditions and modify its control parameters accordingly.

2.3 Feedback-controlled and closed-loop drug delivery

Feedback-controlled drug delivery systems introduce an additional layer of control by using information obtained from the biological environment to influence subsequent drug administration. In a basic feedback system, a physiological variable is measured and compared with a desired target, reference value, or predefined range. The difference between the measured value and the target is then used to determine an appropriate control action, such as increasing, decreasing, or maintaining drug delivery. Closed-loop insulin delivery is one of the clearest clinical examples of this concept. In such systems, continuous or repeated glucose measurements provide information that can be used to adjust insulin administration rather than relying entirely on a fixed dosing schedule. Similar control concepts have also been explored in areas such as neuromodulation, bioelectronic medicine, implantable sensors, and sensor–actuator-based therapeutic systems. The major advantage of feedback control is that the treatment decision is linked to the current state of the biological system. This allows the delivery process to respond to fluctuations that would not be captured by a fixed release formulation. However, feedback control alone does not necessarily mean that the system is learning or calibrating itself. A conventional feedback controller may operate continuously while using a fixed target, sensitivity, gain, algorithm, or control rule established during system design. This distinction is important when discussing adaptive drug delivery. A feedback system can respond to changing conditions without necessarily changing the way it interprets those conditions. In contrast, a self-calibrating system should have some ability to use accumulated information from the biological environment to update an internal parameter or response relationship. Therefore, feedback provides the mechanism for responding to biological information, whereas self-calibration introduces an additional process through which the system can adjust its own operating parameters based on observed data.

2.4 Defining self-calibration in drug delivery

The term self-calibrating is sometimes used broadly to describe drug delivery systems that are responsive, sensor-enabled, intelligent, autonomous, or capable of feedback. However, these terms do not necessarily describe the same level of adaptability. A system that detects a physiological stimulus and produces a predetermined response is stimulus-responsive, but it may not be self-calibrating. Similarly, a system that continuously measures a physiological variable and adjusts drug delivery according to a fixed control algorithm may be considered feedback-controlled without necessarily performing calibration. For the purposes of this review, a drug delivery system should be considered self-calibrating only when there is evidence that the system can obtain information from the biological environment and use that information to modify an internal

parameter or operating rule. At least four elements should therefore be demonstrable: (i) an initialization, adaptation, or learning phase following exposure to the relevant biological environment; (ii) estimation of a local or patient-specific baseline, response relationship, sensitivity, or other system parameter; (iii) modification of a release threshold, sensitivity, gain, actuation rule, or an equivalent control parameter using the acquired information; and (iv) subsequent drug-delivery behaviour based on the updated parameter set. This definition also implies that calibration should have a measurable effect on system performance. Simply collecting physiological data is not sufficient if those data do not influence subsequent operation. Likewise, a system should not be classified as self-calibrating merely because it can respond to a stimulus or because its release rate changes over time because of passive material degradation. Evidence of an actual parameter update or an equivalent change in the control strategy is required. An experimentally useful way to distinguish self-calibration from conventional feedback is therefore to ask whether the system behaves differently after calibration than it would have behaved using its original settings. For example, if exposure to an individual's physiological environment leads the system to estimate a new baseline and subsequently changes the release threshold or control sensitivity, this would provide stronger evidence of self-calibrating behaviour. In contrast, if the system simply follows the same predefined response function throughout treatment, it would be more appropriately classified as stimulus-responsive or feedback-controlled. This distinction is particularly relevant for personalized drug delivery. Biological conditions can vary considerably between individuals and may also change within the same individual over time. A self-calibrating system could, in principle, account for some of this variability by updating its operating parameters rather than depending entirely on population-level assumptions established during formulation development. However, the presence of an adaptive algorithm or sensor alone should not be taken as evidence of self-calibration. The calibration process, the parameter being updated, the data used for the update, and the resulting change in delivery behaviour should all be clearly demonstrated.

Table 1. A practical taxonomy of adaptive drug-delivery systems

Generation	Sensing	Baseline estimation	Threshold adaptation	Feedback	Interpretation
G0: Conventional	No	No	No	No	Pre-programmed release
G1: Stimuli-responsive	Implicit/direct	Usually no	Usually no	Limited	Release depends on a predefined stimulus-response relation

G2: Sensor-triggered	Yes	Usually no	Limited	Partial	Measured signal triggers an action
G3: Closed loop	Yes	Target/reference	May occur	Yes	Output is repeatedly adjusted using feedback
G4: Self-calibrating	Yes	Yes	Yes	Yes	Post-exposure information changes operating parameters
G5: Predictive/autonomous	Yes	Continuous	Continuous	Yes	Adaptive control integrates longitudinal prediction and uncertainty

3. Biological Initialization: Why Tissue Exposure Matters

The initial period after administration or implantation can be considered an initialization window rather than simply a transition to steady-state operation. During this period, the delivery system encounters the actual biological environment, including proteins, cells, extracellular matrix components, ionic gradients, metabolites, oxygen levels, and inflammatory mediators. These interactions can alter the physical and biochemical properties of the device. Protein adsorption, formation of a protein corona, biofouling, and foreign-body responses may influence sensor sensitivity, mass transport, and drug-release behaviour. These changes create two important calibration challenges. First, the measured signal may differ from the signal obtained during laboratory calibration because the biological environment can affect sensor performance. Second, the relationship between a biomarker and the required therapeutic dose may vary among individuals or disease states.

Therefore, calibration can be considered at two levels: sensor calibration, which establishes the relationship between the measured signal and the corresponding biological quantity, and therapeutic calibration, which determines how the measured biological state should be translated into an appropriate drug-release action. A self-calibrating system should ideally be capable of updating one or both relationships based on information obtained after exposure to the biological environment.

4. Sensing and Signal Processing

Self-calibration begins with reliable measurement of the biological environment. Enzymatic sensors, electrochemical electrodes, optical probes, aptamer-based recognition elements, and multi-analyte sensor arrays can provide information about glucose, pH, oxygen, cytokines, enzymes, redox state, and other physiological variables. Single-analyte sensing can simplify system design and control, whereas multi-analyte sensing may provide a more complete picture of complex biological conditions. Signal processing is an important part of this process because raw sensor signals may contain noise, drift, or transient disturbances. Filtering, outlier rejection, drift correction, sensor fusion, and uncertainty estimation can help convert these signals into a more reliable representation of the local biological state. During initialization, methods such as moving averages, adaptive filters, or model-based estimation may be used to establish a baseline while reducing the influence of temporary changes, such as the inflammatory response following implantation. Multi-sensor systems can further improve reliability through redundancy and cross-validation. When multiple sensors provide consistent information, confidence in the estimated physiological state can increase. In contrast, significant disagreement between sensors may indicate sensor malfunction or environmental interference and can be used to trigger a safe fault state rather than an inappropriate drug-release response. This feature is particularly relevant for long-term delivery systems, where gradual sensor drift and changes in the biological environment are difficult to avoid.

5. Adaptive Control and Release Threshold Adjustment

The controller converts information obtained from biological sensing into a therapeutic action. Depending on the design and complexity of the delivery system, different control strategies may be applied. Classical proportional-integral-derivative (PID) control is attractive because its behaviour is relatively interpretable and can be tuned using established control principles. Model predictive control (MPC) can incorporate system constraints and predicted future states, making it potentially useful when drug transport or release involves time delays. Bayesian approaches can represent uncertainty, and update parameter estimates as new observations become available. Machine-learning methods may also be useful for identifying nonlinear relationships between biological signals and therapeutic responses when sufficiently representative data are available. However, the choice of control algorithm alone does not establish that a system is self-calibrating. The important feature is whether the system can update a parameter that controls the relationship between the sensed biological state and the therapeutic response. Such a parameter may include a concentration threshold, controller gain, release sensitivity, permeability setting, degradation trigger, actuator duty cycle, or another variable that has a measurable effect on drug release. The updated parameter should remain relevant to subsequent operation rather than being limited to a temporary response to a single stimulus. An important consideration is distinguishing adaptive control from simple stimulus responsiveness. A stimulus-responsive system may change its release when the biological environment changes, but it may continue to follow the same predefined response relationship. In contrast, a self-calibrating system modifies that relationship using information obtained during operation. This distinction provides a more practical basis for evaluating whether a delivery platform is genuinely adaptive. A useful experimental design is therefore to compare three conditions: an uncalibrated system, the same system after initial tissue exposure and calibration, and a control system that receives similar biological stimuli but is not allowed to update its parameters. Differences in release behavior, accuracy, stability, or therapeutic performance between these groups can provide stronger evidence that calibration contributes to

the observed improvement rather than simply demonstrating a conventional stimulus-response effect.

6. Materials and Actuation Mechanisms

Adaptive control requires an actuator or responsive material that can translate a control decision into a measurable change in drug release. Different material and device platforms can provide this function. Hydrogels can regulate drug diffusion by changing their swelling, mesh size, or permeability in response to environmental conditions. Biodegradable polymers can modify release as polymer erosion or degradation progresses, while nanoparticles can alter drug exposure or release through changes in surface chemistry, molecular interactions, or cleavage of drug-linking groups. Microfluidic and electronically controlled devices can provide more direct regulation by controlling reservoirs, valves, pumps, or other release components. For self-calibrating systems, the material response should be considered together with the sensing and control strategy. A material that responds too slowly may introduce a delay between the control decision and the actual change in drug release. Similarly, irreversible or poorly reproducible material responses can make repeated calibration difficult. Changes in material properties during long-term use may also affect the relationship between the controller output and the resulting drug-release rate. The opposite problem can occur when a material responds very rapidly. If the controller is not appropriately tuned, rapid changes in release can produce overshoot, fluctuations, or unstable behaviour. Therefore, material selection should consider response speed, reversibility, stability, reproducibility, and compatibility with the intended control algorithm. Overall, a self-calibrating delivery platform should be viewed as a coupled sensor–controller–material–drug system rather than a collection of independent components. The sensing layer determines the biological state, the controller decides the required response, and the material or actuator converts that decision into drug release. Effective self-calibration therefore depends on the compatibility and coordinated performance of all these components.

7. Quantitative Metrics for Self-Calibration

One of the major needs in the field is a standardized framework for evaluating adaptive performance. Conventional drug-release studies commonly report cumulative drug release as a function of time, release rate, or kinetic parameters. Although these measurements are important, they do not by themselves demonstrate that a system has calibrated or adapted to its biological environment. Additional quantitative metrics are therefore required to determine whether calibration produces a measurable improvement in system performance.

The following parameters can be used to characterize different aspects of self-calibration:

- Calibration error: Difference between the biological baseline or system parameter estimated by the device and an independently validated reference value. Lower calibration error indicates more accurate estimation.

- **Threshold shift:** Change in the release threshold or other control parameter before and after the initialization or calibration phase. A measurable shift can provide direct evidence of parameter adaptation.
- **Adaptation gain:** Change in therapeutic output produced per unit change in the relevant biological state after calibration. This indicates how the system's sensitivity changes following adaptation.
- **Response latency:** Time between detection of a relevant physiological change and the appearance of a measurable therapeutic response. Shorter and consistent latency may be desirable for rapidly changing conditions.
- **Drift rate:** Change in sensor output, sensitivity, or control threshold over time when the biological or reference condition remains stable. This metric is particularly important for long-term implanted systems.
- **Recalibration efficiency:** Degree to which system performance returns toward the desired target after deliberate perturbation, environmental change, or sensor drift. It provides an indication of whether the system can recover from changes during operation.
- **Specificity and false-trigger rate:** Frequency with which non-target signals or unrelated biological changes produce an unintended drug-release response. These parameters are important for minimizing inappropriate dosing.
- **Hysteresis:** Difference in system response when the same stimulus is approached through increasing versus decreasing trajectories. Significant hysteresis may affect the accuracy of adaptive control.
- **Reproducibility:** Variability of calibrated parameters across different devices, manufacturing batches, and biological subjects. Good reproducibility is necessary for reliable translation and manufacturing.
- **Longitudinal stability:** Ability of the system to retain its calibrated performance during extended operation. This metric is especially relevant for long-term delivery platforms where material degradation, sensor drift, and biological changes may occur.

Together, these metrics provide a broader assessment of self-calibration than conventional release profiles alone. They can help distinguish a system that simply responds to a stimulus from one that estimates, updates, and maintains its operating parameters in response to changes in the biological environment.

Table 2. Suggested minimum evidence package for claiming self-calibration

Evidence level	Required observation	Why it matters	Preferred comparator
1. Sensing	Signal tracks the relevant tissue variable	Establishes measurement validity	Reference analytical method
2. Initialization	Post-exposure baseline differs from pre-exposure calibration were expected	Shows biological interaction is considered	Pre-exposure baseline

3. Parameter update	Threshold/gain/actuation parameter changes using measured data	Defines calibration	No-update control
4. Functional benefit	Release or therapeutic output improves after calibration	Links calibration to utility	Uncalibrated system
5. Robustness	Performance persists despite drift or changing conditions	Tests real-world relevance	Static threshold system
6. Translation	Safety, manufacturing and algorithmic behavior are characterized	Supports clinical development	Current standard of care

8. Pharmacokinetic–Pharmacodynamic Modelling

Adaptive drug delivery can influence not only the rate and pattern of drug release but also the relationship between drug exposure and therapeutic response. Pharmacokinetic–pharmacodynamic (PK/PD) modeling can therefore provide an important link between sensor measurements, drug concentrations, and clinical outcomes. Pharmacokinetic models describe how a drug is absorbed, distributed, metabolized, and eliminated, while pharmacodynamic models describe how drug exposure is related to the resulting therapeutic or biological effect. Physiologically based pharmacokinetic (PBPK) models can incorporate information about tissue distribution, organ function, and physiological changes, which may be particularly useful for adaptive delivery systems operating under variable biological conditions. Patient-specific observations can also be incorporated into these models to improve the estimation of drug exposure and response. Bayesian updating provides one possible approach for combining prior information with newly collected patient-specific data and progressively refining model parameters. For self-calibrating delivery systems, it is important that models account for uncertainty rather than producing only a single predicted value. Biological measurements may contain noise, and the estimated relationship between biomarker levels and therapeutic response may change over time. A controller should therefore be able to recognize uncertainty and apply appropriate safety limits rather than responding aggressively to an unreliable signal. Model validation should consequently involve more than fitting a model to the original dataset. Calibration data should be assessed against independent datasets, and sensitivity analysis should be used to identify parameters that strongly influence therapeutic outcomes. Stress testing under biologically plausible disturbances, such as sensor drift, temporary signal loss, or changes in drug clearance, can further determine whether the model and controller remain stable under realistic conditions. Such modelling approaches can support the development of safer and more individualized self-calibrating drug-delivery systems.

9.1 In vitro evaluation

In vitro evaluation of self-calibrating drug-delivery systems should extend beyond conventional static buffer release studies. Although standard dissolution testing is useful for understanding basic release of kinetics, it may not adequately represent the changing biological conditions encountered after administration or implantation. Dynamic experimental systems can expose delivery platforms to controlled changes in glucose, pH, enzyme concentration, inflammatory mediators, oxygen levels, or other relevant physiological signals. Serum-containing or protein-rich media can be used to investigate the effects of protein adsorption and changes in material or sensor performance. Repeated challenge-and-recovery cycles can further assess whether the system maintains its response after multiple biological stimuli and whether it can recover following temporary disturbances or drift. Microfluidic platforms and bioreactor-based models can introduce controlled flow, concentration gradients, and mass-transport conditions that more closely approximate the physiological environment. Importantly, in vitro evaluation, an initial calibration phase followed by a deliberate biological perturbation. Measurements should determine whether the system updates its operating threshold, sensitivity, or release behavior after calibration and whether this adaptation is reproducible across repeated experiments. Appropriate controls should also be included to distinguish true self-calibration from a predefined stimulus-response effect.

9.2 In vivo evaluation

In vivo models provide the biological complexity required to evaluate self-calibrating drug-delivery systems under conditions that cannot be fully reproduced in vitro. These studies can provide information on tissue interactions, pharmacokinetics, immune responses, sensor performance, drug distribution, and therapeutic outcomes. Diabetes models are particularly useful for evaluating glucose-responsive systems because glucose levels can be experimentally varied, and clinically relevant to therapeutic endpoints can be measured. Tumor models can be used to investigate microenvironment-responsive delivery, while wound-healing models may help evaluate systems responding to inflammation or other local biological changes. Long-duration studies are especially important for self-calibrating systems because calibration performance may change with prolonged exposure to the biological environment. Sensor drift, material degradation, foreign-body responses, changes in tissue properties, and fluctuations in physiological conditions may develop over weeks or months. Therefore, short-term efficacy alone may not provide sufficient evidence of stable adaptive performance.

Translation from animal models to humans remains challenging because species differ in drug metabolism, immune responses, tissue architecture, biomarker concentrations, and physiological dynamics. These differences can influence both sensing accuracy and therapeutic response. Consequently, vivo studies should report not only drug-delivery efficacy but also calibration stability, parameter drift, device failures, adverse biological responses, and the frequency of recalibration events. Such information is important for

determining whether the adaptive behavior observed in experimental models can be maintained safely during longer-term clinical use.

10. Disease-Specific Applications

10.1 Diabetes and metabolic disease

Diabetes represents one of the most developed areas for adaptive drug delivery because glucose is a measurable physiological variable that is directly related to insulin requirements. Glucose-responsive systems and closed-loop insulin delivery therefore provide an important conceptual pathway toward more adaptive therapy. Continuous glucose monitoring combined with automated insulin administration demonstrates how real-time sensing can be used to adjust treatment according to changing glucose levels. However, conventional closed-loop control should be distinguished from self-calibration. A system may automatically adjust insulin delivery without necessarily updating its underlying response parameters based on patient-specific observations. Future self-calibrating approaches could potentially account for changes in insulin sensitivity or individual metabolic responses over time.

10.2 Oncology

Tumor microenvironments contain several characteristics that can be exploited for responsive drug delivery, including acidic pH, hypoxia, altered enzyme activity, and redox imbalance. Responsive nanoparticles and other smart carriers can use these local features to promote drug release preferentially within tumor tissue. [16] Such approaches may improve spatial control of drug exposure and reduce unnecessary release in healthy tissues. The more challenging step is developing systems that can adapt to the biological variability of individual tumors. Tumor properties can differ between patients and may also change during disease progression or treatment. A self-calibrating system would therefore need to identify relevant features of the local tumor environment and adjust its release response accordingly, rather than simply following a predefined trigger-response relationship.

10.3 Wound healing and inflammatory disease

The biological environment of a wound changes continuously throughout the healing process. Changes in inflammatory mediators, enzyme activity, pH, oxidative stress, and tissue composition can provide potential signals for responsive drug delivery. Responsive hydrogels and biosensor-integrated materials can therefore serve as useful platforms for investigating adaptive therapeutic strategies. A self-calibrating system could potentially establish the baseline characteristics of an individual's wound and modify drug release as inflammation decreases or increases. This may be particularly useful when prolonged or repeated treatment is required. However, reliable calibration would need to account for

changes in the wound environment as well as possible for sensor drift and material degradation during healing.

10.4 Neurological and bioelectronic applications

Closed-loop neuromodulation provides an example of how sensing and therapeutic actuation can be connected through a physiological endpoint. Similar architectures could potentially be combined with pharmacological delivery, particularly when neural activity or local biochemical signals provide information about disease state. Such systems may allow therapy to be adjusted according to real-time changes rather than following a fixed treatment schedule. However, neurological applications introduce additional challenges, including signal latency, energy consumption, long-term electrode stability, device reliability, cybersecurity, and algorithmic validation. These factors become particularly important when sensing and drug delivery are integrated into an implantable system that may operate for extended periods.

10.5 Infection and cardiovascular disease

Infection-responsive delivery systems can exploit pathogen-associated enzymes, changes in pH, inflammatory signals, or biofilm-related characteristics to trigger localized drug release. Such approaches may provide a means of increasing antimicrobial exposure at infected sites while limiting unnecessary systemic exposure. Cardiovascular applications may similarly use thrombus-associated signals, local biochemical markers, or changes in hemodynamic conditions to regulate therapeutic delivery.

These applications demonstrate the broad potential of adaptive drug-delivery technologies, but their level of development varies considerably.

Some systems have demonstrated stimulus-responsive or feedback-controlled behavior, whereas others remain largely conceptual. Therefore, evidence should clearly distinguish demonstrated self-calibrating systems from responsive technologies that could potentially be adapted for self-calibration. This distinction is essential when assessing the current maturity and clinical potential of the field.

11. Artificial Intelligence: Opportunity and Constraint

Artificial intelligence (AI) and machine-learning methods may provide useful tools for improving self-calibrating drug-delivery systems. They can assist with physiological state estimation, multivariable signal fusion, detection of complex patterns, and prediction of changing therapeutic requirements. Temporal or recurrent models can capture relationships that develop over time, while reinforcement learning can be used to formulate drug delivery as a sequential decision-making problem in which treatment actions are adjusted according to observed outcomes. However, the use of an AI algorithm does not by itself demonstrate

that a delivery system has achieved meaningful therapeutic adaptation. Machine-learning models depend strongly on the quality and representativeness of their training data. Differences between training conditions and real biological environments can lead to distribution shifts and unexpected model behavior. External validation, uncertainty assessment, and testing under clinically relevant disturbances are therefore important before such models are used for therapeutic control. For safety-critical drug delivery applications, hybrid control architectures may provide a more practical approach. Mechanistic models and rule-based safety limits can constrain the actions of a learning algorithm, while machine learning can be used for tasks such as pattern recognition, signal interpretation, or parameter estimation. This combination can help retain predictable safety boundaries while allowing some degree of adaptation. Other considerations, including cybersecurity, data integrity, interpretability, and continuous monitoring of model performance, should also be treated as part of the therapeutic system itself. Because biological conditions and device performance may change over time, AI-based controllers may require periodic evaluation and validation throughout their operational lifecycle. Thus, AI should be viewed as an enabling technology for self-calibration rather than as evidence of self-calibration on its own.

12. Translational, Regulatory and Ethical Considerations

The translation of self-calibrating drug-delivery systems into clinical practice presents challenges that are different from those associated with conventional fixed-release formulations. A system that can modify its behavior after administration may not have a single, completely fixed performance profile. Manufacturing variation, sensor drift, changes in the biological environment, software modifications, and patient-specific learning can all influence how the system operates and, ultimately, how much drug is delivered. For clinical translation, clear and traceable calibration procedures will be important. Systems should operate within defined safety limits and should be capable of entering a fail-safe state when sensor signals become unreliable or when unexpected conditions are detected. Other important requirements may include auditability of algorithmic decisions, validated software-development processes, cybersecurity protections, long-term biocompatibility, and monitoring of device performance after clinical deployment. Attention may also be required for software or algorithm updates that could change therapeutic behavior after approval. Ethical considerations are equally important because self-calibrating systems may make treatment adjustments that are not directly determined by a clinician at every step. Patients should be appropriately informed when a device has the ability to modify its therapeutic behavior based on collected data. Clinicians should also have access to understandable information about the system's operating state, relevant parameter changes, and the factors that contributed to a dosing decision. Overall, successful translation will require more than demonstrating improved drug-release performance. Safety, reproducibility, transparency, cybersecurity, clinical validation, and accountability should be considered together during system development.

Establishing these requirements early may help ensure that the benefits of adaptive delivery do not come at the cost of reduced control or uncertainty in clinical use.

13. Research Gaps and Future Roadmap

Despite rapid progress in responsive drug-delivery technologies, several important gaps must be addressed before self-calibrating systems can achieve wider clinical translation. The field would benefit from greater consistency in terminology, experimental methods, quantitative evaluation, and long-term validation. The following areas represent key priorities for future research:

- Establish a consensus definition of self-calibration: A clear definition should distinguish true self-calibration from general stimulus responsiveness, feedback control, and autonomous operation.
- Standardize calibration protocols: Studies should report how calibration is performed, which parameters are updated, and the magnitude of threshold or sensitivity changes before and after calibration.
- Develop benchmark datasets and reference environments: Common datasets and well-characterized tissue-mimicking environments could allow different sensing and control approaches to be compared under similar conditions.
- Integrating biological sources of drift into system design: Sensor drift, biofouling, material degradation, and foreign-body responses should be considered as part of the calibration and control problem rather than treated only as independent failure modes.
- Combine mechanistic and data-driven modelling: PK/PD models can provide biological constraints, while machine-learning methods can capture complex relationships. Explicit uncertainty estimation should be included to improve safety during uncertain conditions.
- Perform long-duration in vivo studies: Extended studies are needed to evaluate calibration stability, parameter drift, device reliability, and biological responses rather than focusing only on short-term therapeutic efficacy.
- Develop integrated manufacturing specifications: Sensors, responsive materials, microfluidic components, electronics, and control software should be considered as a single system during manufacturing and quality assessment.
- Establish regulatory frameworks for adaptive systems: Regulatory approaches should address adaptive software and combination products while ensuring bounded, traceable, and auditable system behavior.
- Prioritize clinically meaningful endpoints: Adaptive systems should ultimately be compared with current standards of care and clinically relevant outcomes, rather than only with laboratory controls or conventional release formulations.
- Develop privacy-preserving patient-specific learning: Long-term adaptive therapy may benefit from patient-specific data while maintaining appropriate privacy and

data-security protections. Learning architectures should therefore be designed to personalize treatment without unnecessarily exposing sensitive patient information.

Addressing these gaps will require collaboration between pharmaceutical scientists, materials researchers, engineers, clinicians, data scientists, and regulatory experts. Progress should focus not only on making delivery systems more responsive, but on demonstrating that calibration produces measurable, reproducible, safe, and clinically meaningful improvements over existing approaches.

15. Conclusion

Self-calibrating drug delivery should be viewed as a specific systems-engineering concept within the broader field of intelligent and responsive therapeutics. Its defining feature is not simply the ability to detect a biological stimulus or adjust drug release, but the use of post-exposure biological information to establish or update operating parameters that influence subsequent drug delivery. This distinction separates self-calibration from systems that only respond to predefined stimuli or repeatedly follow a fixed feedback rule.

The field already contains many of the technologies required to develop such systems, including biosensors, responsive polymers, nanocarriers, microfluidics, bioelectronics, closed-loop control, PK/PD modelling, and machine-learning methods. [1–4] The major opportunity now lies in integrating these technologies into a coordinated platform in which sensing, calibration, control, and drug release are directly connected. Importantly, the value of calibration should be demonstrated through measurable and reproducible improvements in therapeutic performance. Future progress will depend not only on increasing technological complexity but on establishing strong experimental evidence that a system can reliably sense its biological environment, estimate relevant parameters and uncertainty, safely adjust its operating conditions, and maintain its performance during long-term use. Standardized definitions, quantitative metrics, long-duration validation, and appropriate regulatory frameworks will be essential for meaningful comparison between emerging platforms. Ultimately, self-calibrating drug delivery has the potential to move therapeutic systems beyond fixed and broadly predefined treatment strategies toward patient-adaptive delivery that responds to individual biological conditions over time. A rigorous systems-level framework can help transform this concept from a broad technological aspiration into a quantitatively validated and clinically useful approach to precision therapy.